

Big Data1 Final Project

This project is in two parts that should be done in a group.

1. The first part of this project is to identify a dataset relevant to your interests (e.g., finance, healthcare, e-commerce, social media, IoT, etc.) and demonstrate the use of data by applying the different topics that we have used to date to show a business result – this could be in the form of a report or a series of graphical representations that gives insights into the business activities of that organization.
2. The second part of this project is to prepare a power point slide presentation to record a video where you will be given a 20-30 min window to present the business and what it does and what your analysis found by explaining the results and why you used the different technologies in the process.

Project Tasks

1. Dataset Selection and Description:

- Search for and select a dataset that aligns with your interests or career goals. Potential sources include:

Open data platforms (e.g., Kaggle, Data.gov, UCI Machine Learning Repository).

- Provide a brief overview of the dataset, including:
 - Data size and format (structured, semi-structured, unstructured).
 - Key features/columns.
 - Business questions you aim to answer.

2.Data Storage and Preprocessing:

- Store the dataset in **HDFS** and/or **MongoDB**, depending on its structure.
- Perform preprocessing tasks using **Pig** for raw data cleaning and aggregation.

3. Data Analysis:

- Use **Hive** to analyze the structured parts of the dataset.
- Write at least three queries to answer specific questions based on the dataset.

4. ETL Pipeline Design (SSIS):

- Create an ETL process using **SSIS** to:
 - Extract data from HDFS, MongoDB, or other sources.

- Transform the data to align with the chosen business questions.
- Load the cleaned and processed data into **SQL Server** for further analysis.
- Write an SQL query in SQL Server to verify and display the loaded data.

5. Advanced Analytics (SSAS):

- Design an OLAP cube in **SSAS** to analyze multidimensional data. Examples include:
 - Trends over time.
 - Geographic or demographic segmentation.
 - Custom metrics specific to the dataset.

Report:

The expected deliverables for this project include a detailed overview of the chosen dataset, including its description and the business questions it aims to address. You should implement a functional big data pipeline, showcasing the ability to process and store data using tools such as HDFS, Hive, and Pig, as well as integrate semi-structured or unstructured data with MongoDB. The project should include a fully operational ETL pipeline designed in SSIS, with a demonstration of data extraction, transformation, and loading into SQL Server, accompanied by a query verifying the data's presence in SQL Server. You can also design a multidimensional OLAP cube in SSAS for advanced data analysis and provide a report. Comprehensive documentation of the process and a professional presentation of the insights and outcomes, and a video recording demonstrating the execution and functionality of the project components should also be included.

All group members must turn on their video during the presentation and actively participate by presenting a specific part of the project, demonstrating their individual contributions and understanding of the work.

Project Mark Breakdown

1. Dataset Selection and Relevance (10%)
2. Usage of the Hadoop components (20%)
3. Pipeline Implementation (30%)
4. Data Transformation and Analysis (10%)
5. Visualization and Reporting (10%)
6. Documentation and Presentation (10%)
7. Originality and Ingenuity (10%)

Example Scenario

There are many uses cases to choose from (Banking, Sales, Manufacturing, Health Care, Pharma - below is an example of a scenario to engage in all aspects of technology that we have touched so far

1. Health Care - Hospital analysis of patient data
 - a. Number of critical, severe, mild accidents in a year
 - b. Create a table that show the Patient with differing columns for example
 - i. A group of patients has name address, accident, severity.
 - ii. Another group of patients has name, accident, severity, medication, addict.
 - c. You could show how many patients were in accidents that were admitted to the hospital as opposed to how many were released the same day based on severity of accidents.
 - d. You could also show this in charts - example a month-to-month breakdown of accidents or show a line graph of accidents in correlation to weather patterns etc. – or the breakdown of age and gender as lines graphs as well